

FE Environmental — Master Cheat Sheet

Cumulative formula & concept reference — grows one topic per study day

Last updated: June 21, 2026 • Topics through Day 5

Mathematics (Day 1)

Algebra

- Quadratic: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$; discriminant $b^2 - 4ac$.
- Logs: $\log(ab) = \log a + \log b$; $\log_b \frac{a}{b} = \log a - \log b$;
 $\log a^n = n \log a$; $\log_b x = \frac{\ln x}{\ln b}$.

Trigonometry

- $\sin^2 \theta + \cos^2 \theta = 1$; $1 + \tan^2 \theta = \sec^2 \theta$.
- $\sin 2\theta = 2 \sin \theta \cos \theta$; $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$.
- Law of sines: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$.
- Law of cosines: $c^2 = a^2 + b^2 - 2ab \cos C$.

Analytic geometry

- Distance $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.
- Line: $y = mx + b$, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Circle: $(x - h)^2 + (y - k)^2 = r^2$.

Calculus

- $\frac{d}{dx} x^n = nx^{n-1}$; $(uv)' = u'v + uv'$; $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$.
- $\frac{d}{dx} \sin x = \cos x$; $\frac{d}{dx} \cos x = -\sin x$; $\frac{d}{dx} e^x = e^x$; $\frac{d}{dx} \ln x = \frac{1}{x}$.
- $\int x^n dx = \frac{x^{n+1}}{n+1}$ ($n \neq -1$); $\int \frac{dx}{x} = \ln |x|$; $\int e^x dx = e^x$.

Vectors & matrices

- $\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}| \cos \theta = A_x B_x + A_y B_y + A_z B_z$.
- $|\mathbf{A} \times \mathbf{B}| = |\mathbf{A}||\mathbf{B}| \sin \theta$.
- 2×2 determinant $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$.

Series & mensuration

- Arithmetic: $S_n = \frac{n}{2}(a_1 + a_n)$. Geometric: $S_n = \frac{a_1(1-r^n)}{1-r}$; $S_\infty = \frac{a_1}{1-r}$ ($|r| < 1$).
- Circle $A = \pi r^2$, $C = 2\pi r$. Sphere $V = \frac{4}{3}\pi r^3$, $A = 4\pi r^2$. Cylinder $V = \pi r^2 h$.

Fluid Mechanics I (Day 2)

Properties

- $\gamma = \rho g$; water $\rho = 1000 \text{ kg/m}^3$, $\gamma = 9.81 \text{ kN/m}^3$.
- SG = $\frac{\rho}{\rho_w} = \frac{\gamma}{\gamma_w}$; kinematic $\nu = \frac{\mu}{\rho}$.
- Newtonian: $\tau = \mu \frac{dv}{dy}$ (linear in velocity gradient).

Statics

- $p = \gamma h$ (gauge); $p_{abs} = p_{atm} + p_{gauge}$; $p_{atm} = 101.3 \text{ kPa} = 10.33 \text{ m H}_2\text{O}$.
- Pressure head $h = \frac{p}{\gamma}$.
- Plane force $F_R = \gamma \bar{h} A$; center of pressure $y_{cp} = \bar{y} + \frac{I_{xc}}{\bar{y}A}$.
- $I_{xc} = \frac{bh^3}{12}$ (rect.), $\frac{\pi D^4}{64}$ (circle).

- Buoyancy $F_B = \gamma_{fluid} V_{disp}$.

Continuity

- $Q = AV$; $A = \frac{\pi}{4} D^2$; $\dot{m} = \rho Q$.
- $A_1 V_1 = A_2 V_2 \Rightarrow V_2 = V_1 \left(\frac{D_1}{D_2} \right)^2$.

Energy / Bernoulli

- $\frac{p_1}{\gamma} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\gamma} + \frac{V_2^2}{2g} + z_2$ (no losses).
- With machines: $+h_{pump} = \dots + h_{turbine} + h_L$.
- Torricelli $v = \sqrt{2gh}$; Pitot $V = \sqrt{2g \Delta h} = \sqrt{\frac{2\Delta p}{\rho}}$.

Reynolds & regime

- $Re = \frac{\rho V D}{\mu} = \frac{VD}{\nu}$; non-circular $D_H = \frac{4A}{P}$.
- Laminar $Re < 2100$; turbulent $Re > 4000$; laminar $f = \frac{64}{Re}$.

Fluid Mechanics II / Pipe Flow (Day 3)

Major (friction) loss

- Darcy–Weisbach: $h_f = f \frac{L}{D} \frac{V^2}{2g}$ (any fluid); $\Delta p = \gamma h_f$.
- $h_f \propto V^2$, $\propto L$, $\propto 1/D$.
- Laminar ($Re < 2100$): $f = \frac{64}{Re}$. Turbulent: Moody diagram via Re , ε/D .
- Colebrook: $\frac{1}{\sqrt{f}} = -2 \log_{10} \left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{Re \sqrt{f}} \right)$ (not in Handbook; use Moody chart).
- Fully rough zone: f depends on ε/D only.

Hazen–Williams (water, turbulent)

- $V = k_1 C R^{0.63} S^{0.54}$; $k_1 = 0.849 \text{ SI}$, 1.318 US .
- $R = \frac{D}{4}$ (full pipe); $S = \frac{h_f}{L}$; higher C = smoother.

Minor losses

- $h_{f,fit} = C \frac{V^2}{2g}$ (Handbook symbol is C , not K); entrance 0.5, exit 1.0, elbow ≈ 0.9 , globe valve ≈ 10 .
- Total $h_L = f \frac{L}{D} \frac{V^2}{2g} + \sum C \frac{V^2}{2g}$.

Pumps

- Fluid power $\dot{W} = \gamma Q H = \rho g Q H$; brake $\dot{W}_b = \frac{\gamma Q H}{\eta}$.
- Operating point = pump curve $\cap$ system curve ($H = H_{stat} + kQ^2$; concept).
- Series: heads add; parallel: flows add (concept, not in Handbook).
- Affinity: $\frac{Q_2}{Q_1} = \frac{N_2}{N_1}$, $\frac{H_2}{H_1} = \left(\frac{N_2}{N_1}\right)^2$, $\frac{P_2}{P_1} = \left(\frac{N_2}{N_1}\right)^3$.
- Avoid cavitation: $\text{NPSH}_A > \text{NPSH}_R$.

Hydraulics — Open-Channel Flow (Day 4)

Geometry

- $R_H = \frac{A}{P}$ (area/wetted perimeter); $D_H = \frac{4A}{P} = 4R_H$; full pipe $R_H = \frac{D}{4}$.

- Hydraulic depth $y_h = \frac{A}{T}$ (T =top width). Rect: $A = by$, $P = b + 2y$. Trap ($z:1$): $A = (b + zy)y$, $P = b + 2y\sqrt{1 + z^2}$.

Manning (uniform flow)

- $Q = \frac{K}{n} A R_H^{2/3} S^{1/2}$, $v = \frac{K}{n} R_H^{2/3} S^{1/2}$; $K = 1.486$ US, 1.0 SI.
- Larger $n \Rightarrow$ lower v , Q ; $Q \propto S^{1/2}$, $R_H^{2/3}$.

Specific energy / critical

- $E = \alpha \frac{v^2}{2g} + y = \alpha \frac{Q^2}{2gA^3} + y$; E min at y_c .
- Critical: $\frac{Q^2}{g} = \frac{A^3}{T}$; rect $y_c = (\frac{q^2}{g})^{1/3}$, $q = Q/B$.
- Alternate depths: same E . $E_{\min} = \frac{3}{2}y_c$ (rect; not in Handbook).

Froude & regime

- $Fr = \frac{v}{\sqrt{gy_h}} = \sqrt{\frac{Q^2 T}{gA^3}}$; $Fr < 1$ sub, $= 1$ crit, > 1 super.

Hydraulic jump

- $y_2 = \frac{y_1}{2} (-1 + \sqrt{1 + 8Fr_1^2})$ (super $\rightarrow$ sub, dissipates energy).
- $M = \frac{Q^2}{gA} + Ah_c$ conserved (sequent depths).

Weirs

- Rect suppressed $Q = CLH^{3/2}$; contracted $Q = C(L - 0.2H)H^{3/2}$; V-notch $Q = CH^{5/2}$.
- C : rect 3.33 US/1.84 SI; 90° V-notch 2.54 US/1.40 SI.

Orifices

- Free: $Q = CA_0\sqrt{2gh}$; submerged: $Q = CA\sqrt{2g(h_1 - h_2)}$; $C = C_c C_v$ (≈ 0.61 sharp).
- Tank drain $\Delta t = \frac{2(A_t/A_0)}{\sqrt{2g}} (h_1^{1/2} - h_2^{1/2})$, $A_t = \frac{\pi D_t^2}{4}$.

Blowers

- $P_w = \frac{WRT_1}{Cne} [(\frac{P_2}{P_1})^{0.283} - 1]$; $C = 550$ US/29.7 SI, $R = 53.3$, $n = 0.283$, e =eff.

Notes

- Partial-flow circular sewer chart: Q_{max} at $d/D \approx 0.93$, $v_{max} \approx 0.81$.
- Flumes/culverts: no rating equation in Handbook (use energy/Manning/orifice).

Surface Water & Hydrology (Day 5)

Rational method

- $Q = CIA$: Q peak discharge (cfs), C runoff coeff, I intensity (in/hr), A area (acres); 1 acre-in/hr $\approx$ 1 cfs.
- Composite $C_w = \frac{\sum C_i A_i}{\sum A_i}$. Storm duration $= t_c$; IDF: longer duration $\Rightarrow$ lower I .
- C tables, t_c & IDF formulas *not* in Handbook (given in problem).

NRCS/SCS curve number

- Runoff depth $Q = \frac{(P - 0.2S)^2}{P + 0.8S}$ (if $P > 0.2S$); P, Q, S in inches.
- $S = \frac{1000}{CN} - 10$; $CN = \frac{1000}{S + 10}$. Higher $CN \Rightarrow$ more runoff.
- Volume[ac-ft] $= \frac{Q[\text{in}]}{12} \times A[\text{ac}]$.

Water budget / evaporation

- $P + Q_{in} - Q_{out} + Q_g - E_s - T_s - I = \Delta S_s$ (inputs - outputs = Δ storage).
- Pan: $E_L = P_c E_p$; $P_c = 0.3 - 0.85$, common 0.7.
- Unit hydrograph: direct runoff from 1 unit excess rain over set duration (linear).

Detention/routing

- Storage $V = \int (Q_{in} - Q_{out}) dt$; constant rates $V = (Q_{in} - Q_{out}) \Delta t$.
- Detention = slow release (cuts peak); retention = permanent pool. Routing algorithms (Muskingum/Puls) *not* in Handbook.

Streeter-Phelps DO sag

- $D = \frac{k_d L_a}{k_r - k_d} [e^{-k_d t} - e^{-k_r t}] + D_a e^{-k_r t}$; $DO = DO_{sat} - D$.
- $t_c = \frac{1}{k_r - k_d} \ln [\frac{k_r}{k_d} (1 - D_a \frac{k_r - k_d}{k_d L_a})]$ (max deficit / min DO).
- k_d deoxygenation, k_r reaeration (base e , d^{-1}); L_a, D_a = mixing-zone values.

BOD / temp / loading

- BOD exerted $BOD_t = L_o (1 - e^{-k_d t})$; remaining $= L_o e^{-k_d t}$.
- $k_T = k_{20} \theta^{T-20}$; θ : BOD 1.135 ($\leq 20^\circ\text{C}$) / 1.056 (21–30), reaeration 1.024.
- Loading $M[\text{lb/day}] = C[\text{mg/L}] Q[\text{MGD}] 8.34$.